

Database Management Systems — Lab Project Proposal

Smart Pharmacy Management System with Expiry & Prescription Tracking

1. Project Title

Smart Pharmacy Management System with Expiry & Prescription Tracking

2. Brief Description

The Smart Pharmacy Management System is a database-driven web application designed to digitize and streamline the daily operations of a retail pharmacy. The system manages medicine inventory, supplier records, customer profiles, prescriptions, and sales transactions through a centralized MySQL database. A key feature is its intelligent alert mechanism, which automatically notifies the pharmacist about medicines nearing their expiry date, low stock levels, and pending prescription refills — issues that are often mishandled in traditional pharmacies. The system also generates real-time sales and inventory reports to support better decision-making.

By integrating Point-of-Sale (POS) billing, prescription tracking, and analytics into a single platform, the project demonstrates practical use of core DBMS concepts such as normalization, joins, triggers, stored procedures, transactions, and views.

3. Modules / Features

Module 1 — Inventory & Supplier Management

- Add, update, and delete medicines with categories, dosage, and unit price
- Manage medicine batches (batch number, manufacturing date, expiry date, quantity)
- Maintain supplier records and purchase orders
- Record stock-in entries with automatic batch creation

Module 2 — Sales & Billing (POS)

- Search medicines by name or barcode
- Add items to cart, apply discounts, calculate tax
- Generate and print invoices
- Auto-deduct stock from the appropriate batch via database trigger
- Maintain complete sales transaction history

Module 3 — Customer & Prescription Management

- Register customer profiles with contact details and medical notes
- Upload and store prescriptions linked to specific customers and doctors
- Maintain doctor records
- Link prescriptions to sales for controlled/prescription-only medicines
- Refill reminders for chronic-medication customers

Module 4 — Alerts, Reports & Admin Dashboard

- Expiry alerts (30, 60, 90-day thresholds) and low-stock alerts
- Daily, weekly, and monthly sales reports
- Top-selling medicines and revenue analytics
- User authentication with role-based access (Admin, Pharmacist, Cashier)
- Activity logs for audit trail

4. Technologies Used

Layer	Technology
Frontend	HTML5, CSS3, JavaScript, Bootstrap
Backend	Python (Flask framework)
Database	MySQL
DB Connector	mysql-connector-python
Templating	Jinja2
Tools	VS Code, MySQL Workbench, Git & GitHub, Postman

DBMS Concepts Demonstrated:

Normalization (up to 3NF), Primary & Foreign Keys, Composite Keys, Joins (Inner, Left, Right), Triggers (auto stock deduction, expiry logging), Stored Procedures (billing, expiry check), Views (low-stock view, expiring-soon view), Transactions (billing rollback on failure), and Indexing.

5. Group Members & Roles

#	Member Name	Role	Module / Responsibilities
1	Zohaib Aziz	Team Lead	Inventory & Supplier Management — schema design for medicines, batches, suppliers; CRUD operations; stock-in workflows; overall project coordination.
2	Abdullah Tariq	Developer	Sales & Billing (POS) — cart logic, invoice generation, stock-deduction triggers, transaction handling.
3	Fahad Hussain Soomro	Developer	Customer & Prescription Management — customer/doctor records, prescription linking, refill reminders.
4	Haris Bin Arif	Developer	Alerts, Reports & Admin Dashboard — expiry/low-stock alerts, sales reports, user authentication, role-based access.

All members will jointly contribute to database schema design, ER diagram preparation, testing, and the final presentation to ensure equal participation throughout the project lifecycle.

6. Expected Outcome

A fully functional, web-based pharmacy management system that handles real pharmacy workflows end-to-end — from purchasing stock and managing prescriptions to billing customers and generating reports — while clearly demonstrating mastery of relational database concepts learned throughout the course.